Exp No:15 Experiment to understand the data preprocessing in Data science

AIM:

To understand the data preprocessing in data science.

CODE:

```
from google.colab import files
files.upload()
import numpy as np
import pandas as pd

# Load dataset
df = pd.read_csv("/content/pre-process_datasample.csv")
print(df)

# Display information
df.info()

# Find mode of Country
print(df.Country.mode())
print(df.Country.mode()[0])
print(type(df.Country.mode()))

# Fill missing values
df.Country.fillna(df.Country.mode()[0], inplace=True)
df.Age.fillna(df.Age.median(), inplace=True)
df.Salary.fillna(round(df.Salary.mean()), inplace=True)

print(df)

# One-hot encoding of Country
print(pd.get_dummies(df.Country))

# Create updated dataset
updated_dataset = pd.concat(
   [pd.get_dummies(df.Country), df.iloc[:, [1, 2, 3]]],
   axis=1
)

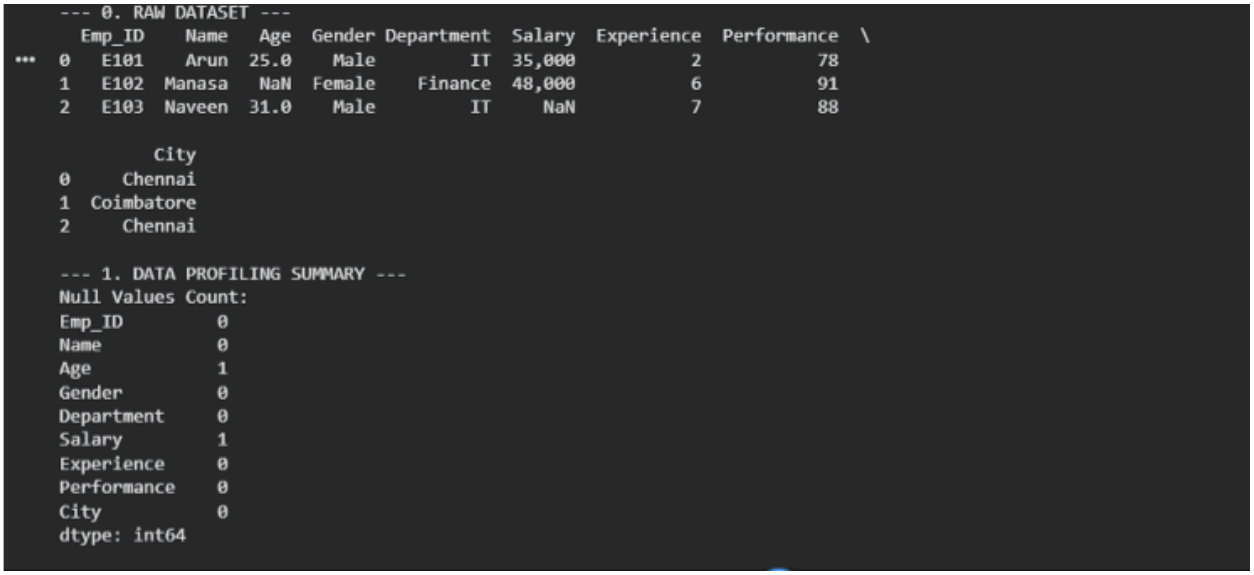
print(updated_dataset)

# Convert Purchased values to 0 and 1
updated_dataset.Purchased.replace(
   ['No', 'Yes'],
```

```
    [0, 1],
    inplace=True
)

print(updated_dataset)
```

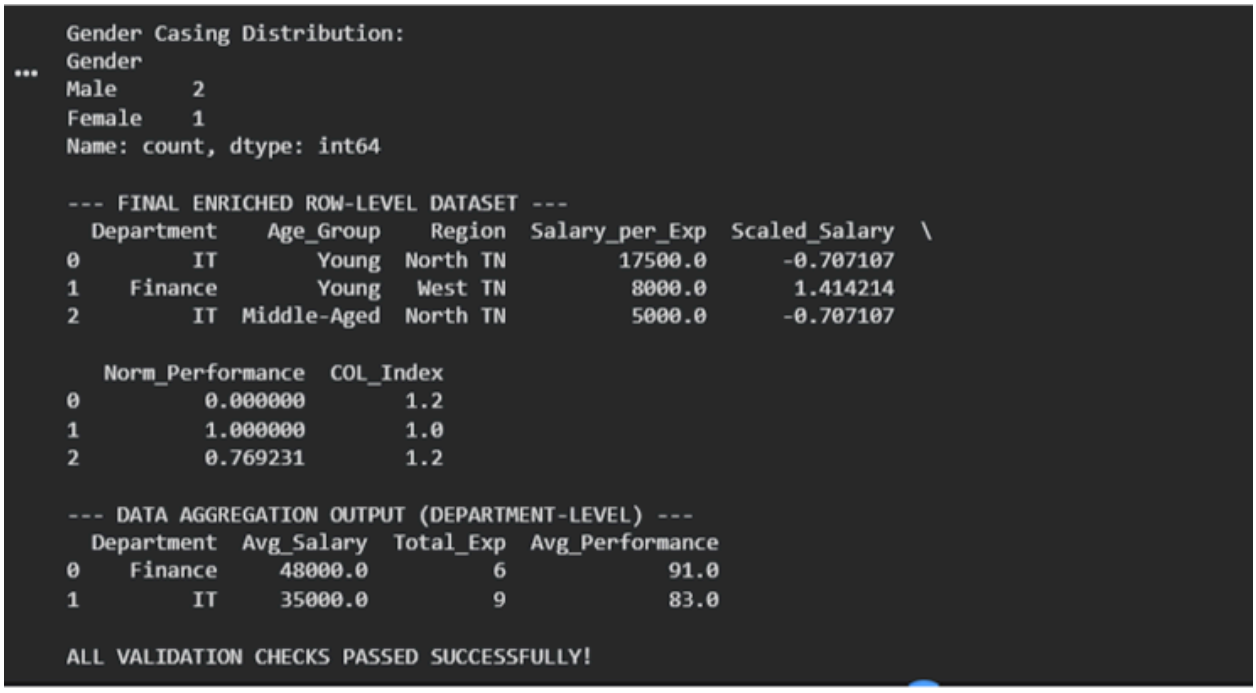
OUTPUT:



```
--- 0. RAW DATASET ---
  Emp_ID    Name   Age  Gender Department  Salary  Experience  Performance  \
0   E101    Arun  25.0    Male         IT  35,000           2           78
1   E102  Manasa   NaN  Female    Finance  48,000           6           91
2   E103  Naveen  31.0    Male         IT     NaN           7           88

         City
0     Chennai
1  Coimbatore
2     Chennai

--- 1. DATA PROFILING SUMMARY ---
Null Values Count:
Emp_ID         0
Name           0
Age            1
Gender         0
Department     0
Salary         1
Experience     0
Performance    0
City           0
dtype: int64
```



```
Gender Casing Distribution:
Gender
Male      2
Female    1
Name: count, dtype: int64

--- FINAL ENRICHED ROW-LEVEL DATASET ---
  Department    Age_Group    Region  Salary_per_Exp  Scaled_Salary  \
0         IT        Young  North TN         17500.0      -0.707107
1    Finance        Young   West TN          8000.0       1.414214
2         IT  Middle-Aged  North TN          5000.0      -0.707107

   Norm_Performance  COL_Index
0          0.000000        1.2
1          1.000000        1.0
2          0.769231        1.2

--- DATA AGGREGATION OUTPUT (DEPARTMENT-LEVEL) ---
  Department  Avg_Salary  Total_Exp  Avg_Performance
0    Finance     48000.0          6             91.0
1         IT     35000.0          9             83.0

ALL VALIDATION CHECKS PASSED SUCCESSFULLY!
```

RESULT:

Thus,the given data was analyzed successfully and required output was obtained.